

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that: **port 53 is unreachable** when attempting to resolve the domain name yummyrecipesforme.com.

This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message: "**udp port 53 unreachable**"

The port noted in the error message is used for: Domain Name System (DNS) traffic

The most likely issue is: The DNS server at 203.0.113.2 is offline, the DNS service has crashed, or a firewall configuration is blocking UDP traffic on port 53.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time incident occurred: The incident was observed over a multi-minute span starting at **13:24:32**, with repeated automated attempts recorded at **13:26:32** and **13:28:32**.

Explain how the IT team became aware of the incident: The IT team became aware of the incident after a user reported that they could not load or connect to the website yummyrecipesforme.com via their web browser.

Explain the actions taken by the IT department to investigate the incident: The network security team responded by running a network protocol analyzer tool, **tcpdump**, to capture traffic between the local computer (192.51.100.15) and the DNS server (203.0.113.2) during a connection attempt.

Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.): The resulting logs revealed that the local computer sent three separate UDP queries to port 53 on the DNS server (203.0.113.2) requesting the IP address for yummyrecipesforme.com. Each request was immediately met with an ICMP error message from the server indicating that UDP port 53 is unreachable. Because the DNS query failed, the browser could not obtain the IP address needed to initiate an HTTPS

connection to the web server.

Note a likely cause of the incident: The network security team suspects the primary cause is that the DNS service on server 203.0.113.2 is stopped/disabled, or a network firewall rule was incorrectly modified to block incoming traffic to port 53. The DNS server might be down due to a successful Denial of Service attack or a misconfiguration.